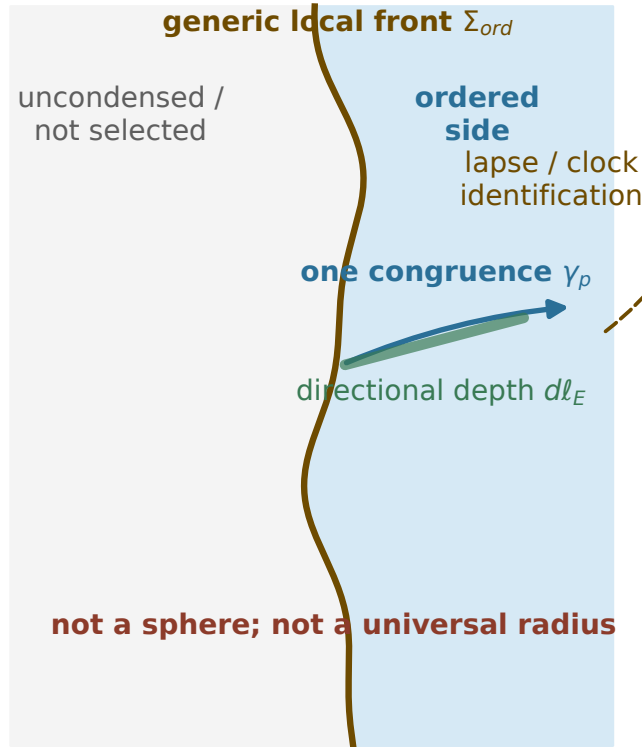


Shape-agnostic external Euclidean view and internal cosmological history

external Euclidean description

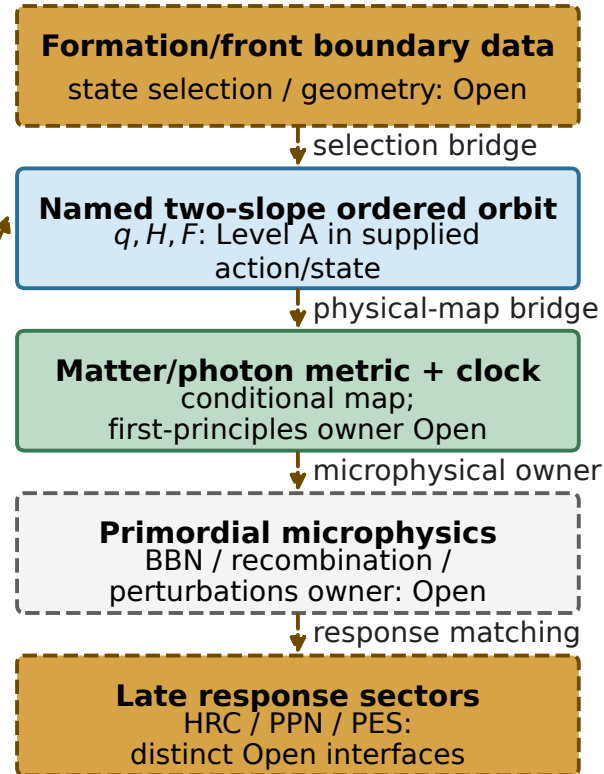


shape and global topology are not specified

Directional internal/external identities

$$d\tau = (N_\Phi/c_u)d\ell_E \quad H = (c_u/N_\Phi)a^{-1}da/d\ell_E$$

internal history and ownership



owned inside named completion
 conditional identity
 Open bridge/input
 external/imported layer